A Research Program on Collatz Synchronization Dynamics

Final Summary: Papers A–P

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Abstract

This document summarizes a research program on the Collatz conjecture. The program develops three components and a negative result: (A–E) exact mixing theory for the 2-adic Haar kernel; (F–M) local cascade geometry and one-shot obstruction; (N–O) Hensel-chain reformulation; (P) proof that mixing alone cannot resolve Collatz. The central open question is: what deterministic structure prevents an actual Collatz orbit from traversing the infinite one-shot Hensel ray?

1 Four Components

A–E: Mixing theory. The 2-adic Haar Markov kernel P_K on odd residues mod 2^K mixes to Haar uniform in exactly $K-1$ steps (Theorem TMix, Paper B). The exact TV formula is proved. Convergence of the Collatz orbit is equivalent to $\sup_K R_K < \infty$ (Theorem SE, Paper E).

F–M: Local cascade geometry. The valuation process $a_t = v_2(n_t - 1)$ obeys exact algebraic laws within cascades: $a_t \geq 3 \Rightarrow a_{t+1} = a_t - 2$ and $q_{t+1} = 3q_t$ (Theorems VD, OC). Jump heights are characterized by exact congruence conditions (Theorems JE, JE2). After any height $K \geq 4$ jump, the next jump has height $< K$ (Theorem SR), and the phase cannot return to its pre-jump value within the cascade (Theorem UPI). Consequently, height records are strictly increasing and each $K \geq 4$ is achieved at most once (Theorem Monotonicity).

N–O: Hensel-chain structure. The gate-2 jump formula is established (Theorem JE2, Paper N). Phase transport is exact within cascades but not across renewals (scope correction, Paper N). If infinitely many gate-1 records occur at heights $K_i \to \infty$, then $q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$ (Theorem HMC, Paper O). The 2-adic escape pair reformulation: the Collatz conjecture reduces to the nonexistence of an orbit with $q_{\tau_i} \to -3^{-1}$ (2-adic) and $q_{\tau_i} \to +\infty$ (Archimedean).

P: Negative result on mixing. In the infinite Haar renewal model, every Hensel level K is hit infinitely often a.s. (Theorem IH, BC Lemma 2). Therefore: mixing *increases* accessibility to Hensel targets; it does not suppress them. The statistical mixing theory (Papers A–E) is *insufficient* to prove Collatz. The obstruction, if it exists, must arise from the *deterministic* one-shot structure of actual Collatz orbits.

2 Complete Theorem List

Theorem	Paper	Status	Content
ETV	B	Proved	Exact TV formula for P_K
TMix	B	Proved	Mixing time $= K-1$, exact
DR	B	Proved	Dual representation of P_K
SE	E	Proved	Convergence $\iff \sup R_K < \infty$
VD	F	Proved, exact	$a_t \geq 3 \Rightarrow a_{t+1} = a_t - 2$
JH	F	Proved (Haar)	Jump height $\sim$ Geom(1/2)
OC	G	Proved, exact	$a_t \geq 3 \Rightarrow q_{t+1} = 3q_t$
OE	G	Proved	Cascade multiplies q by $3^{\lfloor (k-2)/2 \rfloor}$
JE	H	Proved, exact	Gate-1: height K iff $q \equiv r_K \pmod{2^K}$
Hensel	I	Proved	$r_{K'} \equiv r_K \pmod{2^K}$ for $K' \geq K$
SR	I	Proved, exact	Next jump $< K$ after height K
MC	J	Proved (cond.)	Waiting cost $\Omega(2^K)$
PC	K	Proved	$A(K) = \{1, \ldots, H_K\}$, $H_K = v_2(3^{L_K} - 1)$
PG	L	Proved	Phase semigroup $\mathcal{S}_K = \{0\}$ or $\mathbb{Z}/2^{K-2}\mathbb{Z}$
UPI	M	Proved	Phase frozen within cascade after height K
Monotonicity	M	Proved	Height records strictly increasing; each K one-shot
JE2	N	Proved, exact	Gate-2 jump formula and target $r_{K,s}$
HMC	O	Proved	Gate-1 records $\Rightarrow q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$
IH	P	Proved	Haar model: every K hit infinitely often a.s. (BC2)

Note on conditional results. Papers J–M use a phase transport assumption ($q_{\tau_i} = 3^{S_i} q_{\tau_1}$) that is exact within a single cascade but fails across distinct renewals. Their longer-range conclusions (MC, PG phase theory) are conditional on this assumption. Theorems SR and UPI are exact within their stated cascade-level scope.

3 The Two-Layer Structure

The program reveals that Collatz renewal events split into two types:

- **Generic renewals**: equidistribute rapidly to Haar under P_K (Papers A–E). Height K target hit with probability $2^{-(K-1)}$ per renewal.
- **Record renewals**: lie on the Hensel chain $q \equiv r_K \pmod{2^K}$ (Papers N–O). Form a strictly increasing one-shot sequence (Paper M).

The Haar model conflates these two types: in i.i.d. Haar renewals, every level K is revisited infinitely often. Actual Collatz separates them sharply: records are one-shot, non-revisitable.

4 The Open Problem

Open Problem 1 (One-shot Hensel traversal)**.** Does any Collatz orbit traverse the Hensel chain

$$q_{\tau_1} \equiv r_{K_1}, \quad q_{\tau_2} \equiv r_{K_2}, \quad \dots \pmod{2^{K_i}}, \quad K_1 < K_2 < \cdots \to \infty$$

infinitely often, with each height K_i used exactly once, while $q_{\tau_i} \to +\infty$ in $\mathbb{R}$?

The Collatz conjecture asserts: no. The obstruction must be *deterministic*—arising from the orbit's rigid structure beyond generic mixing.

Equivalent formulation (Paper O). No Collatz orbit has a 2-adic escape pair: a sequence with $q_{\tau_i} \to -3^{-1}$ in $\mathbb{Z}_2$ and $q_{\tau_i} \to +\infty$ in $\mathbb{R}$.

Why this is hard. The Hensel chain conditions are algebraically consistent (Paper O). The Haar model supports infinite traversal (Paper P). The one-shot constraint (Paper M) is the only known structural obstruction, but it does not by itself prevent all infinite sequences—it only requires each level be used at most once. Whether the deterministic Syracuse dynamics can thread an infinite one-shot sequence through the Hensel chain remains open.

5 Honest Assessment

What is new and rigorous. The exact cascade algebra (VD, OC, JE, JE2), the immediate isolation theorem (SR), the phase freezing (UPI), the Hensel chain identification (HMC), and the negative mixing result (IH) appear to be new contributions. The precise separation of local/global, exact/conditional, and mixing/deterministic is new.

What remains open. The Collatz conjecture. The transfer from Haar mixing to actual orbits. The conditional phase theory (Papers J–M) under the cascade transport assumption.

The shape of the problem, after Papers A–P. A divergent orbit would be a "real escape trajectory shadowing a 2-adic Hensel ray"—growing Archimedean while converging 2-adically to -3^{-1}, with each level of the Hensel chain touched exactly once. Collatz asserts this trajectory does not exist. Proving it requires understanding deterministic orbit geometry beyond what statistical mixing captures.